HealthStat

Case Study: Analyzing Healthcare Data in Power BI

Elective Hip Replacement Surgical Impatient Stays: New York State

Project Scope:

- Identify efficiency trends, outliers, and key drivers of length of stay and cost per discharge.
- Build an interactive, branded dashboard using DAX to enable benchmarking and actionable insights into hospital performance.

HealthStat

Home | LOS Comparison | Cost Comparison | Hospital Profile

health_service_area

All

627 Total Surgeons

2.65 Average LOS Days

151 Total Hospitals

26K Total Discharges

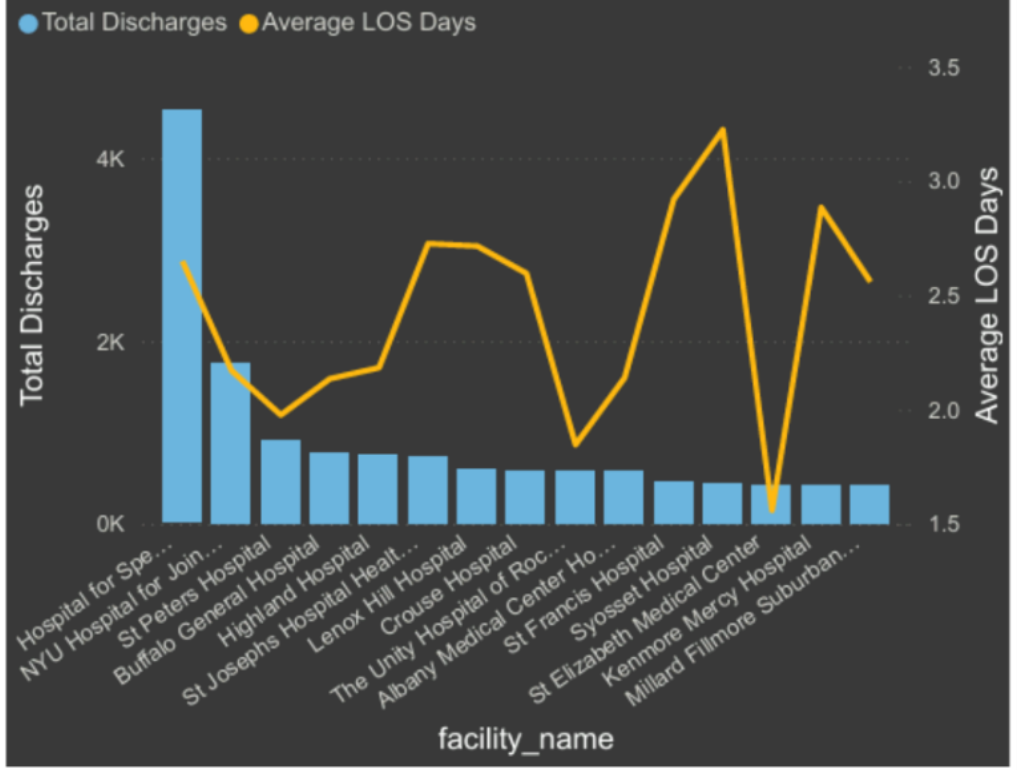
Key influencers | Top segments

What influences Average LOS Days to: Increase ?

When...	....the average of Average LOS Days increases by
apr_severity_of_illness_d... is Extreme	8.96
apr_risk_of_mortality is Extreme	6.09
apr_risk_of_mortality is Major	2.48
health_service_area is New York City	1.68

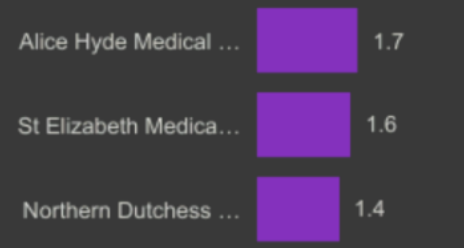
Sort by: Impact Count

Average LOS Days and Total Discharges by top 15 Hospitals



Top 3: Highest Avg LOS Days

Bottom 3: Lowest Avg LOS Days



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health_service_area

All

2.65	$21K	151	26K
Average LOS Days	Average Cost per Dis...	Total Hospitals	Total Discharges

Key influencers | Top segments

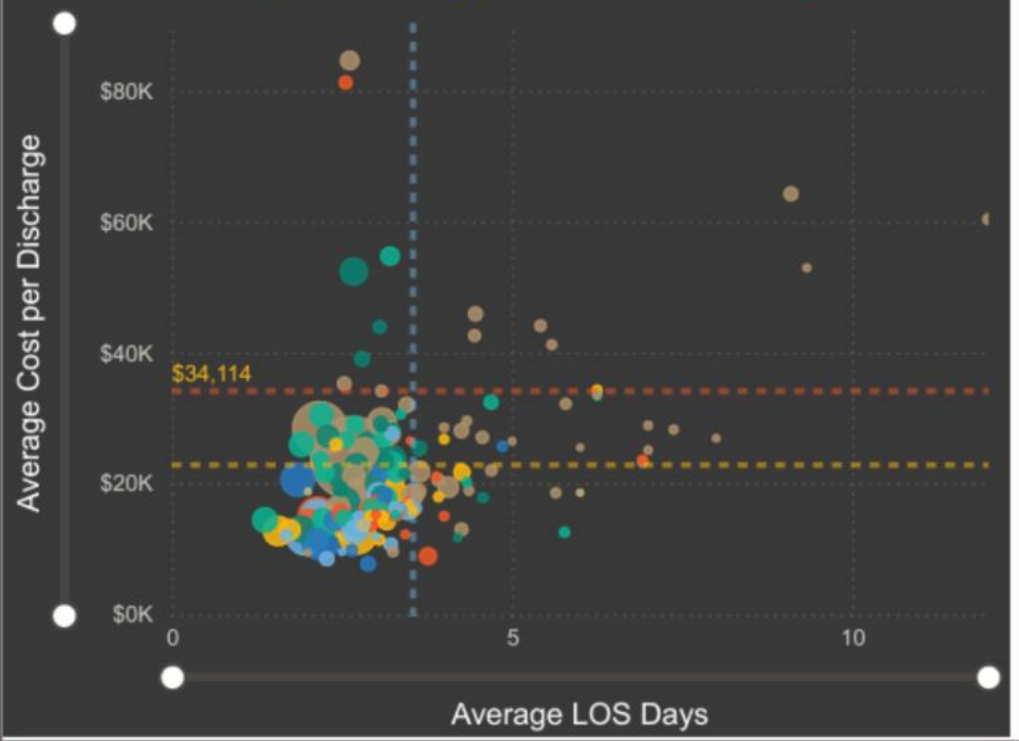
What influences Average Cost per Discharge to Decrease ?

When... ....the average of Average Cost per Discharge decreases by

- health_service_area is Capital/Adiron → $9.33K
- health_service_area is Southern Tier → $8.58K
- health_service_area is Central NY → $7.83K
- health_service_area is Finger Lakes → $7.78K

Sort by: Impact Count

Average LOS vs Average Cost per Discharge by Hospital



Top 3: Highest Avg Cost per Discharge

Bottom 3: Lowest Avg Cost per Discharge

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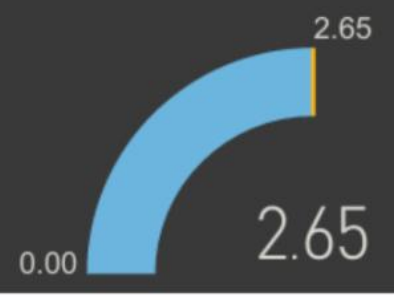
facility_name

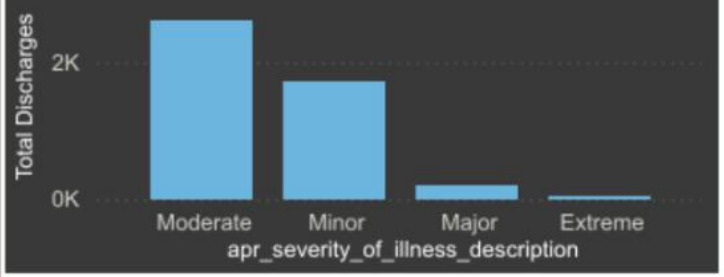
Hospital for Special Surgery

Hospital Profile: Hospital for Special Surgery

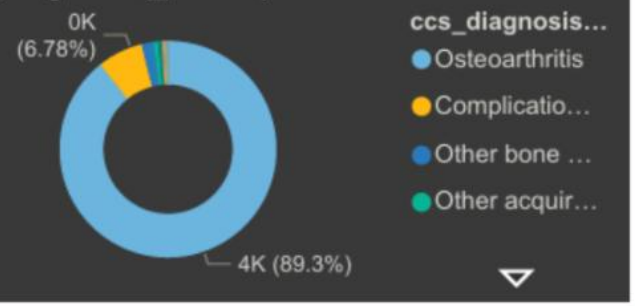
Average LOS Days and Average LOS Days ALL



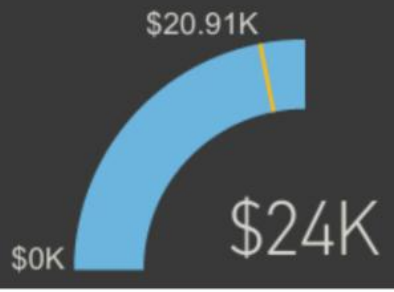
Total Discharges by apr_severity_of_illness_description



Total Discharges by ccs_diagnosis_description

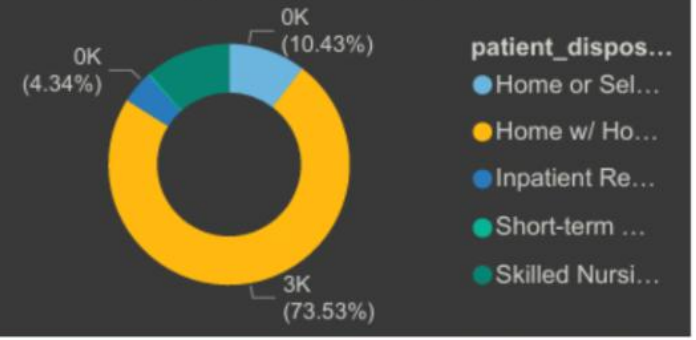


Average Cost per Discharge and Average Cost per Discharge ALL



Total Discharges by apr_risk_of_mortality

Total Discharges by patient_disposition



Key Business Insights:

- 26,000 total discharges across 151 hospitals statewide, with an average length of stay of 2.65 days
- Wide LOS variation across hospitals: Kings County Hospital averages 12.0 days vs. Northern Dutchess at just 1.4 days — an 8x gap signaling major efficiency differences
- Severity of illness is the #1 driver of longer stays, "Extreme" severity increases average LOS by nearly 9 days, followed closely by extreme mortality risk
- NYC hospitals trend longer - Health service area (New York City) independently adds ~1.7 days to average LOS, suggesting regional care complexity or resource constraints
- Cost per discharge averages $21K but ranges dramatically, NYU Lutheran tops the chart at $85K vs. lowest performers near $7–8K
- Geography drives cost: Capital/Adirondack region saves ~$9K per discharge vs. average; rural/upstate regions consistently outperform urban on cost
- Osteoarthritis dominates as the primary diagnosis (89.3% of discharges at Hospital for Special Surgery), confirming this is largely a planned, elective procedure population
- 73% of patients discharge home — high self-sufficiency post-surgery, but 10%+ require inpatient rehab or skilled nursing, representing an opportunity to reduce downstream costs
- Hospitals with high discharge volume do NOT always have the longest stays — volume and LOS are not correlated, meaning high-volume hospitals can still be highly efficient